# HybSeq or GBS for herbarium specimens?

# A battle of reduced representation datasets.

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## Background & Aims

Reduced representation sequencing techniques provide a cost-effective way to obtain datasets for phylogenomic reconstructions. However, these techniques frequently fail when working with old material like herbarium specimens. Factors like DNA degradation and methylation can lead to a loss of restriction sites. Here, we aim to answer the question, which of two prominent reduced representation sequencing techniques, Genotyping by Sequencing (GBS) or HybSeq, is able to retrieve more data from herbarium specimens. In addition, we test the applicability of the Angiosperm353 probe set to *Veronica* subgenus *Pseudolysimachium*, a group proving to be especially challenging for DNA extraction and sequencing due to its high content of secondary metabolites.

## Methods

DNA was extracted from 60 silica-dried and 39 herbarium specimens. HybSeq was performed using the Angiosperm353 probe set in combination with a custom probe set. Adapter removal and quality trimming for both datasets was done using Trim Galore. GBS reads were assembled using ipyrad v0.9.92. Target sequence capture for the HybSeq dataset was done with HybPiper v2.1.6. The multiple sequence alignment for the HybSeq dataset was performed using mafft v7.520. Phylogenetic trees were constructed with IQ-TREE v2.2.5 and assembly statistics were visualized with R v4.3.1.

## Results




**Fig. 1** Recovery heatmap combining percentage of sequenced probe length for Angiosperm353 targets and custom targets. Darker colors indicate higher percentage length recovery, relative to mean target file references.

- The number of retrieved HybSeq reads and loci from herbarium material is **not significantly lower** than from silica material.
- The number of retrieved GBS reads and loci from herbarium material is **significantly lower** than from silica material.
- Differences in phylogenetic trees are present, especially among *V. spicata* and the putative hybrid *V. gryniana*. However, most major clades were resolved in both datasets, with comparable support values.

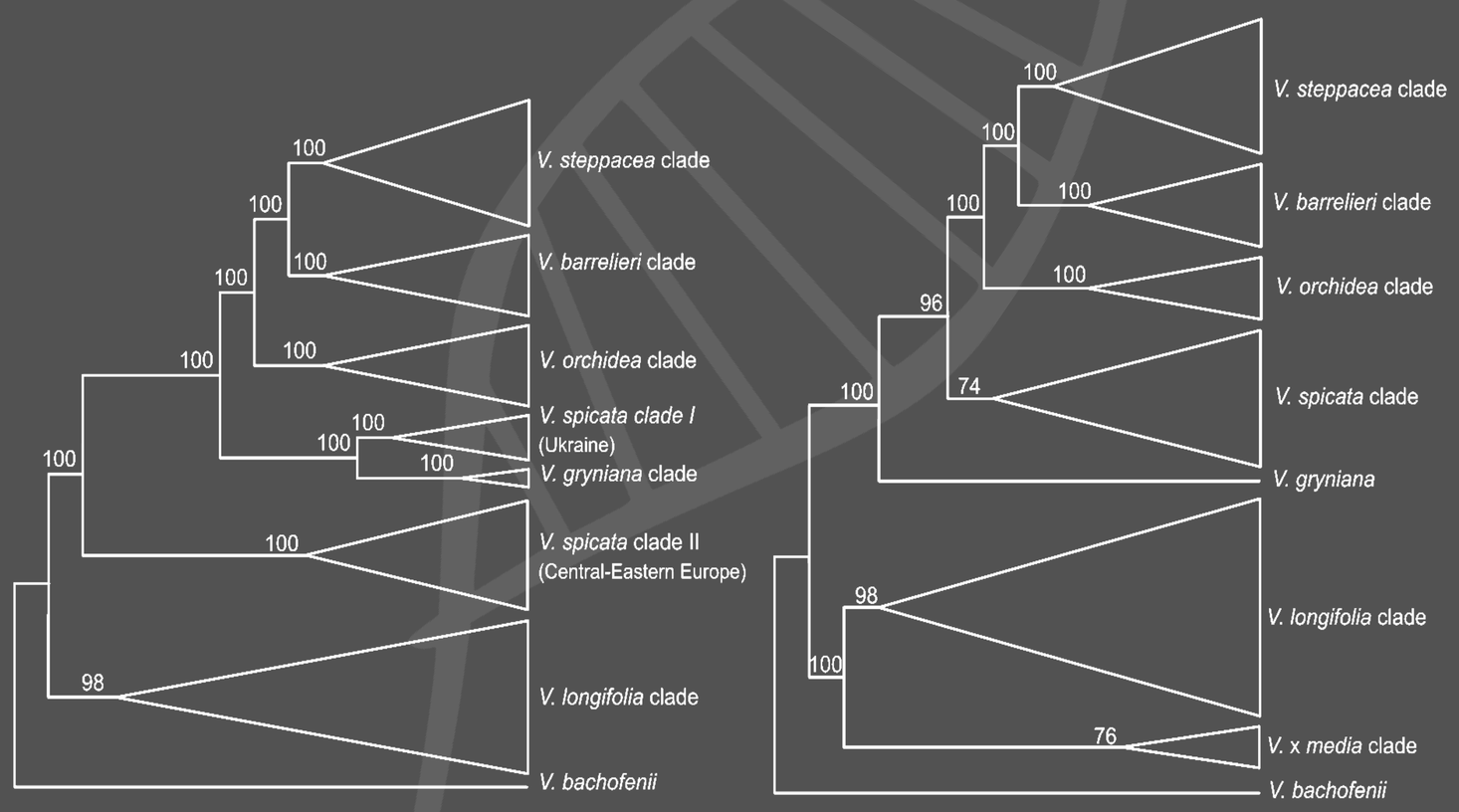



**Fig. 4** Phylogenetic trees (simplified) retrieved from GBS data (left) and HybSeq data (right). Numbers above branches show likelihood bootstrap support [%].

- The percentage length recovery is generally higher in the custom target set than in the Angiosperm353 dataset.

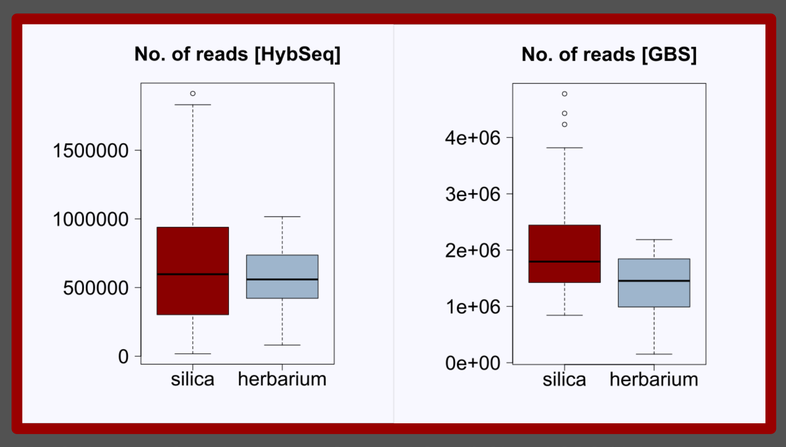



**Fig. 2** Number of reads obtained from HybSeq (left) and GBS (right). Red plots show reads retrieved from silica-dried material, grey plots show reads retrieved from herbarium material.

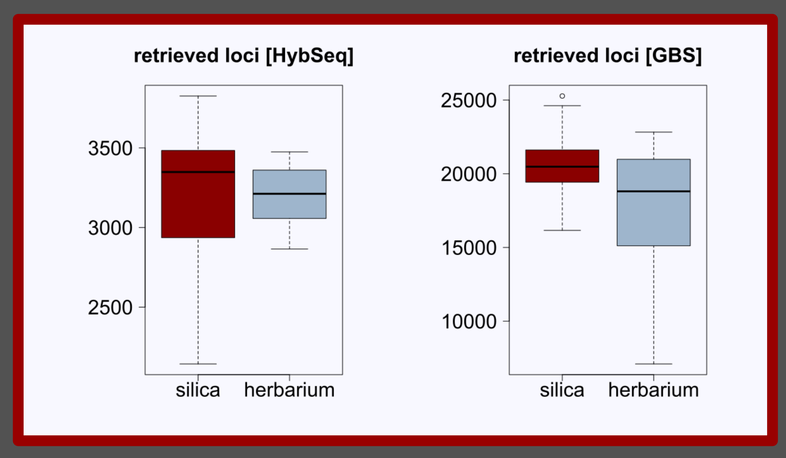



**Fig. 3** Number of retrieved loci from HybSeq (left) and GBS (right). Red plots show reads retrieved from silica-dried material, grey plots show reads retrieved from herbarium material.

## Conclusion

- HybSeq is the better approach for sequencing herbarium material, since no significant differences in retrieved number of reads and loci could be observed between silica material and herbarium material.
- Genotyping by Sequencing is not recommended for herbarium material, since we obtained a significantly lower amount of data and loci.
- Relying on the Angiosperm353 probe set alone is not recommended (for *Veronica).*




Funded by
DFG Deutsche Forschungsgemeinschaft German Research Foundation
VolkswagenStiftung